

7	(a) each wavelength is associated with a discrete <u>change</u> in energy discrete energy <u>change</u> / difference implies discrete levels	M1 A1	[2]
	(b) (i) 1. arrow from -0.54 eV to -0.85 eV, labelled L	B1	[1]
	2. arrow from -0.54 eV to -3.4 eV, labelled S <i>(two correct arrows, but only one label – allow 2 marks)</i> <i>(two correct arrows, but no labels – allow 1 mark)</i>	B1	[1]
	(ii) $E = hc / \lambda$ $(3.4 - 0.54) \times 1.6 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.0 \times 10^8) / \lambda$ $\lambda = 4.35 \times 10^{-7}$ m	C1 C1 A1	[3]
	(c) $-1.50 \rightarrow -3.4 = 1.9$ eV $-0.85 \rightarrow -3.4 = 2.55$ eV (allow 2.6 eV) $-0.54 \rightarrow -3.4 = 2.86$ eV (allow 2.9 eV) <i>3 correct, 2 marks with –1 mark for each additional energy</i> <i>2 correct, 1 mark but no marks if any additional energy differences</i>	B2	[2]

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